

LEGALITY AND VENTURE CAPITAL EXITS*

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LEGALITY AND VENTURE CAPITAL EXITS

Abstract

This paper introduces an analysis of the impact of Legality on the exiting of venture capital investments. We consider a sample of 468 venture-backed companies from 12 Asia-Pacific countries, and these countries' venture capitalists' investments in US-based entrepreneurial firms. The data indicate IPOs are more likely in countries with a higher Legality index. This core result is robust to controls for country-specific stock market capitalization, MSCI market conditions, venture capitalist fund manager skill and fund characteristics, and entrepreneurial firm and transaction characteristics. Although Black and Gilson (1998) speculate on a central connection between active stock markets and active venture capital markets, our data in fact indicate the quality of a country's legal system is much more directly connected to facilitating VC-backed IPO exits than the size of a country's stock market. The data indicate Legality is a central mechanism which mitigates agency problems between outside shareholders and entrepreneurs, thereby fostering the mutual development of IPO markets and venture capital markets.

Keywords: Venture Capital, Exits, IPO, Acquisition, Law and Finance

JEL Classification: G24, G28, G31, G32, G35

1. Introduction

The risks and returns to investing in small private companies and companies with intangible assets are pronounced.¹ This is especially the case in smaller and less developed finance markets such as those in the Asia-Pacific region. The legal rights and protections afforded to entrepreneurs and investors are particularly important in venture capital and private equity markets. Legal structures impact the rights and protections of parties and distribution of financial outcomes under various states of the world.

That law matters in many aspects of finance is widely recognized, with studies showing that legal institutions and rules influence capital structure, dividend payout ratios, repurchasing decisions, and capital market returns.² Much of this literature, however, concentrates on publicly traded companies. The impact of legal institutions and structures on private equity investments (such as those undertaken by venture capitalists) has received less attention.

This paper provides a first-ever law and finance analysis of venture capital dispositions. We consider the impact of the quality of legal environment on the choice between public sale (initial public offerings) and private sale transactions (including acquisitions, secondary sales and buybacks, each defined in section 2).³ In our cross-country empirical analysis, we consider the ‘law and finance variables’ from La Porta *et al.* (1997, 1998, 2000), including the efficiency of the judicial system, the rule of law, corruption, risk of expropriation, risk of contract repudiation, and shareholder rights [hereafter ‘the law and finance factors’]. Taken together, these factors comprise the ‘Legality’ in a country (Berkowitz *et al.*, 2003), which measures the quality of a country’s legal system.

The extension of law and finance into the context of venture capital has previously been explored in a few contexts. In a collection of their pioneering work, Gompers and Lerner (1999) discuss some

¹ A more extensive discussion of this can be found in Lerner (1994a,b), Gompers (1995), Gompers and Lerner (1999, 2001a,b), Lerner and Schoar (2004a), and Denis (2004).

² See, e.g., La Porta *et al.* (1997, 1998, 2000), Berkowitz *et al.* (2003), Himmelberg *et al.* (2002), Klapper and Love (2002); for a survey, see Choi, (2002).

³ In this paper we focus on ‘venture capital’ investments loosely defined as early and expansion stage investments (as defined in the Asia-Pacific www.asiaventure.com and Europe www.evca.com). A few investments are also at the later mezzanine stage (described further in section 3). An earlier version of this paper (available upon request) considered private equity buyouts (on the distinctive characteristic of buyouts generally, see, e.g., Wright *et al.*, 2001). Those transactions were excluded from the current version of this paper as they were specific to one country (Australia) in our paper. Regardless, the exclusion/inclusion of buyouts did not materially affect the results in our sample.

aspects of the impact of US taxation and securities laws on the US venture capital industry. Gilson and Schizer (2003) study tax and US venture capital transaction structure. Black and Gilson (1998) consider the importance of an active securities market, and hence the importance of appropriate securities market regulations. Gompers and Lerner (1998) and Jeng and Wells (2000) study venture capital fundraising and the importance of the institutional environment in the fundraising process.

Previous research has documented international differences in developed venture capital markets (see, e.g., Black and Gilson, 1998; Bascha and Walz, 2001b; Schwienbacher, 2002, 2003; Mayer *et al.*, 2004; Rock, 2001; Cumming and MacIntosh, 2003a,b; Hege *et al.*, 2003). Recent studies have also considered the particular characteristics of emerging venture capital markets (if not generally, then “emerging” at least in the sense that the venture capital market is not very well developed). Most notably, Lerner and Schoar (2004b) study contracts in emerging markets; Gompers *et al.* (2003) study legality and entrepreneurship in Eastern versus Western Europe; Wright *et al.* (2002) study venture capital in India; Lockett and Wright (2002) and Lockett *et al.* (2002) study venture capital in Asia; Songtao (2000) discusses venture capital in Taiwan; Kuroki *et al.* (2000) analyze trends in Japan; Wang and Sims (2001) discuss venture capital in Singapore; Fleming (2002) analyzes venture capital in Australia.

In this paper we extend the literature by providing a cross-country law and finance analysis of venture capital exits (IPOs, acquisitions, secondary sales, buyback and write-offs). The exit process is of central interest because venture capital investments typically do not pay dividends; rather, returns are derived from capital gains upon exit. In fact, a venture capitalist’s decision to invest typically depends on exit potential. Where laws impact exit, they also impact the decision to invest and therefore the entire development of a venture capital market. Prior work has not considered private equity exits in more than one or two countries simultaneously, and hence has been unable to fully address the question of how Legality impacts private equity markets.

This paper considers a new dataset of 468 venture capital-backed companies across 12 Asia-Pacific countries: Australia, China, Hong Kong, India, Indonesia, Malaysia, New Zealand, Philippines, Singapore, South Korea, Taiwan and Thailand. A common characteristic across these Asia-Pacific countries is that their venture capital markets are not well developed, particularly relative to the US. A interesting aspect of this empirical study is that we also consider Asia-Pacific venture capitalists’ investments in US-based entrepreneurial firms. The Asia-Pacific region is of particular interest to introduce this type of research as Legality is very heterogeneous in the region and of significant concern to practitioners (see, e.g., Bassolino, 2002, Zhang, 2002, and Stipp, 2002). Among the 468 investments in

our dataset, 78 companies had been exited by means of an IPO, 200 exited by a private sale (acquisition, secondary sale or buyback), and 55 ended up as write-offs; the remainder 135 companies in the data have not yet been exited. The sample covers the 1989-2001 period. The heterogeneity in the data is useful, as it enables us to consider a broad spectrum of countries, as well as statistical selection effects in regards to exited and unexited investments.

The new principal findings derived from the data and institutional context highlight the central importance of Legality for exiting private equity investments in the Asia-Pacific. With a nonlinear concave (logarithmic) specification, we find that an increase in the Legality index from 20-21 (e.g., similar to the difference between Hong Kong to Australia) increases the probability of an IPO by 1.7%, whereas an increase in Legality from 10-11 (e.g., similar to the difference between the Philippines and Indonesia) increases the probability of an IPO by 3.3%.

In their influential paper, Black and Gilson (1998) theorize (without empirically testing) that there exists a central connection between active stock markets and active venture capital markets. In our paper, by contrast, we provide (to the best of our knowledge) a first direct test of this proposition. Our data in fact indicate the quality of a country's legal system is much more directly connected to facilitating VC-backed IPO exits than the size of a country's stock market. Stock market capitalization is statistically unrelated to the probability of a VC-backed company achieving an IPO in our sample (even though there is a correlation between Legality and the size of a country's stock market, and therefore the *appearance* of a causal connection between active stock markets and active venture capital markets, as identified by Black and Gilson, 1998). Overall, our data are consistent with the view that Legality is a central mechanism which mitigates agency problems between outside shareholders and entrepreneurs, thereby fostering the mutual development of IPO markets and venture capital markets.

Our new dataset is the first to be able to measure the effect of Legality on private equity exits. While we focus on Legality, we nevertheless control for a number of other variables that could account for IPO versus other types of venture capitalist exits. These controls include market conditions (such as the country-specific MSCI indices, dummy variables for the Internet bubble, and the country-specific stock market capitalization), country-specific dummy variables, VC fund characteristics (such as limited partnership VCs versus captive VCs, fund capital, fund manager experience, US-affiliated funds, and sector or stage specialization of the fund), and entrepreneurial firm and deal characteristics (including industry market/book ratios, size, syndication, co-investment, investment duration), among other things. Our findings with respect to the importance of Legality on exiting VC investments in the Asia-Pacific

region are robust to all of these control variables. These control variables also highlight some of the more interesting features of VC markets outside of the more heavily studied US market, as discussed herein.

While we control for many factors that we would expect to affect venture capital exits in the countries studied, we also acknowledge that there are limitations in both the breadth and depth to which it was possible to obtain and/or present private confidential information. We discuss the limitations in this paper and suggest avenues for future research.

This paper is organized as follows. In section 2 we describe the institutional environment from which the data are derived, and outline the central hypotheses. In section 3 we introduce the data and provide summary statistics. Section 4 provides a multinomial logit analysis of exit outcomes, and shows the interaction between Legality, country factors and the exit outcomes, with numerous robustness checks. Limitations and future research are discussed in section 5. Conclusions follow.

2. Institutional Structure and Hypotheses

The data (described fully in section 3 below) comprise venture capital exits from Australia, China, Hong Kong, India, Indonesia, Malaysia, New Zealand, Philippines, Singapore, South Korea, Taiwan and Thailand. The data also comprise venture capital exits from Asia-Pacific venture capitalist investments in US-based entrepreneurial firms. In this section we describe the empirical tests that we are able to perform with the venture capital data from these countries: law and finance factors are considered in subsection 2.1; economic, governance and other institutional factors are discussed in subsection 2.2.

2.1. Legality and Venture Capital Exits

There are five principal venture capital exit vehicles (see, e.g., Black and Gilson, 1998 for discussion): (1) initial public offerings (IPOs) – the entrepreneurial firm is listed on a stock exchange for the first time; (2) acquisitions (or ‘trade sales’) – the company is purchased by a larger firm, typically a strategic acquiror, and both the venture capitalist and entrepreneurs sell their interest in the company; (3) secondary sales – the venture capital fund sells its interest, but the entrepreneurs do not sell their interest, to another firm or venture capital fund; (4) buybacks – the entrepreneurs repurchase their interest from the venture capital fund; and (5) write-offs (or ‘liquidation’) – the investors ‘walk away’ from the investment with little or no return. In this paper we introduce the term ‘private exit’ to refer to the group of

acquisitions, secondary sales and buybacks. In some instances in this dataset, we are able to observe when there has been a private exit, but not the specific type of private exit.

Exits have been the subject of much academic venture capital research. Because there is a greater amount of information available for IPO exits (the only ‘public’ exit in that information must be disclosed in a prospectus), IPOs have traditionally been more frequently studied in academic work (see, e.g., Barry *et al.*, 1990, Megginson and Weiss, 1991; Lin and Smith, 1997; Gompers and Lerner, 1999). Recent empirical research has also considered the determinants of exit choice.⁴ There is also a theoretical literature relating convertible securities in venture capital to exit choice (Bascha and Walz, 2001a; Schwienbacher, 2002; see Bergmann and Hege, 1998, for related work on venture contracts). Black and Gilson (1998) discuss the importance of implicit contracts and the importance of IPOs for venture capital markets. The central conjecture in Black and Gilson (1998) is that active stock markets facilitate IPO exits and therefore active venture capital markets. In this paper we empirically test that Black and Gilson conjecture.

Black and Gilson (1998), Schwienbacher (2002), Fleming (2002), Cumming and MacIntosh (2003a,b) and others rank order the exit vehicles by the quality of the entrepreneurial firm (from highest to lowest): IPOs, acquisitions, secondary sales, buybacks and write-offs. One of the many driving principles underlying this ordering (among more than a dozen different factors that could affect the ordering) relates to information asymmetries. In order to maximize the capital gain upon exit, a venture capitalist will choose the exit vehicle for which the new owners are best able to resolve information asymmetry. When informational asymmetries are lowest, the new owners are willing to pay more for the company.⁵ As it is typically most difficult for new owners in an IPO to mitigate informational problems (see Pagano *et al.*, 1998), we may predict that venture capitalists will only take public the best quality firms for which informational asymmetries are least pronounced. In contrast, buybacks involve no informational asymmetries for the ‘new’ owners (the founding entrepreneurs), but do not bring any new capital into the firm (and therefore buybacks rare among firms with significant growth prospects).

⁴ See, e.g., Fleming (2002), Cumming and MacIntosh (2003a, b), Das *et al.* (2003), Schwienbacher (2002). For case studies, see Petty *et al.* (1999). Furthermore, Cochrane (2001), Gompers and Lerner (1999), and Cumming and Walz (2004) provide related analysis on the performance of VC investments.

⁵ The idea that venture capitalists mitigate informational problems and certify the quality of a firm upon going public was first discussed with respect to IPO underpricing by Barry *et al.* (1990) and Megginson and Weiss (1991).

The widely recognized law and finance factors (based on La Porta *et al.*, 1997, 1998, 2000) include civil versus common law systems, the efficiency of the judicial system, the rule of law, corruption, risk of expropriation, risk of contract repudiation, and shareholder rights (in other words, the substantive content of laws pertaining to investing, the quality of their enforcement, and the likelihood that they will need to be enforced). The Legality index is a weighted sum of the factors based on Berkowitz *et al.* (2003), as derived from (La Porta *et al.*, 1997, 1998, 2000). Higher numbers indicate ‘better’ legal systems across each of the factors. Because the rankings are highly collinear, we focus on the Legality index in our regressions to avoid the appearance of data mining. We did consider specific components of the Legality index, but found much similarity in the qualitative results, and therefore focus on the overall Legality index variable.

There are many reasons to believe that Legality matters in the context of venture capital finance. First, all else being equal, a higher Legality index in a country is associated with stronger investor protection, and therefore a more active stock market. Because an active public equity market is essential for a successful venture capital market (Black and Gilson, 1998), we expect countries with a high Legality index to have a more active venture capital market.

Second, we hypothesize that an investor’s sale of their interest in a private entrepreneurial firm is affected by Legality. As mentioned, the exit decision is influenced by (among other things) information asymmetries between the original investor and the new owner(s). We expect Legality to affect the exit strategy because Legality affects the new owner(s) ability to resolve problems resulting from information asymmetries in the sale of the firm (consistent with La Porta *et al.*, 1997, 1998; Shleifer and Wolfenzon, 2002). We predict that Legality will affect the exit vehicles observed across countries. The venture capitalist’s goal is to maximize capital gains upon sale of the entrepreneurial firm. All else being equal, the new investor(s) will pay the most when information asymmetries are lowest. We conjecture that IPOs are less costly exit routes relative to private exits (acquisitions, secondary sales and buybacks) among countries with a higher Legality index and stronger investor protections, and should therefore be observed more frequently in countries with higher Legality indices. Similarly, among countries with very low Legality indices, we would expect to observe a higher frequency of buyback exits because such exits involve the least need for protection for the ‘new’ owners (the entrepreneurs repurchase the venture capitalists’ interest).⁶

⁶ In addition, we also check country-specific factors, as these may also be important in determining the likelihood of a buyback. For example, South Korea made changes to their Commercial Code in 2002 to allow the buyback of shares. We are

2.2. Economic and Governance Factors Affecting Venture Capital Exits

In addition to country legal factors, a variety of other variables may influence the exit outcome of a VC backed company, including market conditions and market capitalization, VC fund and entrepreneur characteristics, and transaction-specific characteristics. We briefly discuss each in turn in this subsection.

Two important types of market variables can affect exit outcomes of VC-backed companies. First, it is well known that favorable stock market returns and bubble periods increase the probability of IPO outcomes, and therefore we control for country-specific MSCI indices, as well as the Internet bubble in our sample. Second, the country-specific stock market capitalization can increase the probability of IPO exits.⁷ Black and Gilson (1998) argue that the development of the public equity market permits VCs to exit through an IPO, which itself is critical to the development of the venture capital market. Thus, it is reasonable to assume that the development of the public equity market is linked to the choice of the exit process for venture capitalists. At the same time, La Porta *et al.* (1997, 1998) document countries with better legal protection of shareholder rights have more developed and bigger equity markets. The exit choice and the development of legal systems may be linked together not because better shareholder protection laws result in lower agency costs between entrepreneurs and outside shareholders and foster IPO activity but because both variables are correlated with the development of stock markets. Therefore, it is important to control for the size of a country's equity market in order to understand the relationship between the VC's exit choice and the legal system.

It is further important to control for VC fund characteristics in assessing VC exit outcomes.⁸ Hsu (2004) shows that entrepreneurial firms chose specific VC's in order to signal their quality and lower the information asymmetries between themselves and market participants. In our empirics, we control for a large number of factors pertaining to VC fund characteristics, including limited partnership VCs versus captive VCs,⁹ fund capital, fund manager experience, US-affiliated funds, and sector or stage

unaware of corporate and securities laws in the other countries prohibiting other exit vehicles in the other countries; country-specific institutional considerations were still considered through the inclusion of country dummy variables in the empirics.

⁷ We owe a special thanks to an anonymous referee for this helpful comment.

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⁹ Captive VCs are expected to have less success, given their comparatively inefficient structures relative to limited partnership VCs (pay per performance sensitivities and internal control decision-making autonomy, among other things; see Gompers and Lerner, 1999, for a discussion and US evidence).

specialization of the fund. We expect these proxies for greater fund sophistication to be positively associated with an increase in IPO exit outcomes.

An entrepreneurial firm's characteristics may likewise affect exit outcomes, as this will be directly pertinent to the degree of information asymmetry faced by the new owners, as well as the success potential and growth options available to the entrepreneur. In our empirical tests we consider a number of entrepreneurial firm characteristics, such as industry market/book ratios, asset intangibility, and size proxies.

Finally, various governance variables, and VC-entrepreneur deal-specific characteristics may impact VC exit outcomes. Our dataset (described in the next section) enables some controls along this dimension, such as syndication and co-investment. Consistent with Gompers and Lerner (1999), we expect syndication to facilitate IPO exits, but not co-investment. Lerner (1994a) shows syndication facilitates deal screening and due diligence, as well as value-added to the entrepreneurial firm. Gompers and Lerner (1996) explain the agency problems associated with co-investment: VC funds from the same VC firm may misallocate their value-added activities, among other things, provided to the entrepreneurial investees in the VC fund's portfolio, to the detriment of institutional investors. Co-investment therefore is likely associated with a reduced probability of an IPO exit and a higher probability of a write-off.

We recognize that there are limitations in the level of detail in the empirical analysis given the extreme difficulties in obtaining reliable private (confidential) venture capital data. For instance, entrepreneur versus VC control over exit may also affect exit choice (Black and Gilson, 1998; Smith, 2000; Kirilenko, 2001; Bascha and Walz, 2001a; Cumming, 2002). Funds in our dataset structured their transactions almost invariably with the use of a VC control in at least some dimension, such as with investor board seats.¹⁰ To the extent that we have information from the funds in the sample, there is little heterogeneity among the security choice and board seats taken for typical investments, and therefore this dimension would not impact upon the exit choice for the VC funds in our sample. We did consider fund effects through the use of fund dummy variables, in case certain funds manifested unobservable

¹⁰ One of the co-authors works for an institutional investor in the Asia-Pacific region, and has access to this information, but is unable to disclose specifics for reasons of confidentiality. Functionally, there are different ways to obtain control, such as by staging, using contractual rights of control, board seats, etc. Our understanding is that the VC funds in the data exercise implicit control, although do not necessarily achieve this control via the same means in all transactions. Further work could investigate the role of different degrees of control on exit outcomes (see also Cumming, 2002, for a European dataset in which control is much more heterogeneous relative to that which is in our current sample). See also the discussion in section 5 below.

differences, but did not find significant differences across different funds. We do discuss the limitation of this detail and others further in section 5 of this paper. The data are first described in section 3, and empirical tests are thereafter provided in section 4.

3. Data and Summary Statistics

3.1. Data Description

This paper analyzes a hand-collected dataset involving 468 venture capital-backed companies from 12 countries in the Asia-Pacific: Australia, China, Hong Kong, India, Indonesia, Malaysia, New Zealand, Philippines, Singapore, South Korea, Taiwan and Thailand. We also study Asia-Pacific venture capitalists' investments in US-based entrepreneurial firms. A common characteristic across these Asia-Pacific countries is that their venture capital markets are not well developed, particularly relative to the US. The data were collected from unsolicited and solicited confidential information memorandum sent by Asia-Pacific VCs to an international fund-of-funds manager (Wilshire Associates, Inc.; <http://www.wilshire.com>). The dataset comprises 53 Asia-Pacific VC funds, but have yet to be judged as suitable for investment or even a due diligence review (thus removing the possibility that only upper quartile VCs are examined). Data were also obtained from captive funds through Wilshire's industry connections. To this extent, we believe that the data are reasonably representative of VCs operating in the Asia-Pacific region, although we do consider possible selection bias. Wilshire's industry connections enabled us to make use of significantly more private information than that which has been previously documented about Asia-Pacific venture capital exits and returns. VCs provide high quality data to prospective investors, and are required to report and validate all of their exits (the data comprise all exited investments from the VCs in the sample, thus avoiding potential biases associated with grandstanding by VCs in the fundraising process; see Gompers and Lerner, 1999, for an analysis of grandstanding).

The variables that we use in the ensuing empirical analyses are described in Table 1. The data are summarized in Table 2 and Figures 1 and 2. Figure 1 indicates the data by year of investment and year of exit. Figure 2 provides a histogram of returns by exit status (IPOs, private exits, and write-offs, as well as by partial realizations¹¹). There also exists investments for which exit has yet to occur. That we have

¹¹ There are 'full' and 'partial' venture capital exits for each exit vehicle (Megginson and Weis, 1991; Gompers and Lerner, 1999; Neus and Walz, 2004; Cumming and MacIntosh, 2003a,b). A full IPO, secondary sale and buyback involves the sale of 100% of the venture capitalist's interest within one year from the date of the first disposition; a partial exit takes more than one year. A full acquisition involves a sale for cash; a partial acquisition involves the receipt of (typically illiquid) shares.

exited, partially exited and unexited investments in our sample is useful for sample selection issues (in the spirit of having as randomized a sample as possible, and in the spirit of using non-exited investments for Heckman sample selection corrections in the multivariate analyses, as described below). The histograms of investment and exit data (Figure 1) and of IRRs (Figure 2) are broadly consistent with other international VC data samples.¹² The frequency of observations for each exit type from each country is indicated in Table 2, alongside the Legality indices. The countries with the greatest representation in our sample are Australia, Taiwan, and Asia-Pacific VC-backed US-based entrepreneurial firms.

[Tables 1 and 2 and Figures 1 and 2 About Here]

The data are similar in size and scope to other hand-collected datasets in academic venture capital research (see, e.g., Lerner and Schoar, 2004b, for a sample of 210 private equity transactions in developing countries). It is difficult to know exactly how representative the exit frequencies are from each of the counties in the dataset because the reporting of VC data from Asia-Pacific markets is in a nascent stage of development, and the extent of coverage is unknown.¹³ We cannot be reasonably certain about aggregate country level data for any of the Asia-Pacific countries,¹⁴ which is in contrast to the US VC market. As such, we are unable to provide comparisons to industry-wide data (as typically done for US-only VC datasets; see Gompers and Lerner, 1999). Our inability to provide reliable industry-wide comparisons on variables of interest is a limitation of most all VC research with international VC samples outside of the US (again, see, e.g., Lerner and Schoar, 2004b).

A full write-off is a complete liquidation of the investment; a partial write-off is a write-down of the book value of the investment. Our concern in this paper is not with investigating the extent of exit; rather, we focus on the preliminary fundamental issue of the exit vehicle outcome. Prior work (e.g., Gompers and Lerner, 1999; Neus and Walz, 2004; Cumming and MacIntosh, 2003a, b), is consistent with the view that the exit vehicle affects the extent of exit, not vice versa. An earlier version of this paper analyzing the extent of exit is available upon request.

¹² See, e.g., Cumming and Walz (2004) for a sample of VC and private equity returns across 39 countries (although Cumming and Walz do not have details on specific exit outcomes, unlike our dataset). Note that the unexited IRRs in our sample are not always at “0.00%”, as unexited IRRs are reported by VC fund managers to their institutional investors. The reporting of unexited IRRs is considered in Cumming and Walz (2004).

¹³ Even in relatively more developed countries in the Asia-Pacific region such as Singapore, the available data from other sources has only recorded the frequency of venture backed IPOs and non-IPO exits (that is, grouping acquisitions, secondary sales, buybacks, and write-offs all in the same category; see Wang and Sims, 2001) and scant other information about the investments and exits.

¹⁴ Australia is in part one exception to this statement (see Cumming *et al.* 2004, for Australian data), but the industry-wide details that are available for the Australian VC market are different from that which we consider in this paper.

The dataset contains information on private equity funds in the Asia-Pacific region that are potentially appropriate for institutional investment, and/or are known to institutional investors (particularly in the case of the captive funds in our sample). To this extent the data set is drawn from funds that are relatively large. If the data are not representative of a particular country, it is probably along the lines of not observing the dispositions from smaller private equity funds unknown to institutional investors from less developed countries (although we do not believe this is the case based on our investigation in conjunction with Wilshire, we offer it as a possible consideration). If at all, this would bias our results in terms of observing less frequently IPO exits in less developed countries. Said differently, if our results pertaining to Legality are skewed due to sample selection bias, they would be skewed in terms of understating the impact of Legality on exits, if we have only the better VC funds from the less developed countries. Our results pertaining to Legality are therefore likely conservative in nature.

The large number of investments and exits in US-based entrepreneurial firms is an interesting aspect of the data (Table 2). *A priori*, these foreign investments were not expected because venture capitalists typically prefer to be geographically proximate to their investees to provide regular value-added advice and more effective governance (Gompers and Lerner, 1999). However, it is an established investment strategy in less developed venture capital markets to relocate some entrepreneurial firms in the portfolio to the US. These companies have the requisite characteristics to attract further rounds of funding (or exit) in the US market, and the venture capitalist is able to arbitrage value differences between low entry prices in the less developed domestic market, and high exit prices in the more developed US market. Below, we compare the performance of these foreign VC investments to the domestic VC investments. We also consider separate subsamples of the data including and excluding the US-exits, as these observations may be unique in regards to the role of legality in affecting exit outcomes.

It is noteworthy that most of the IPOs in the data are from Australia and Taiwan (Table 2). There are a few (4 IPOs of 15 investments) IPOs from New Zealand, and only 9 (from a total of 117 investments) from the US-based entrepreneurs. These exits by Asia-Pacific VCs in US-based entrepreneurial firms may reflect the difficulties in screening and monitoring foreign investments, and the increased difficulty Asia-Pacific venture capitalists have in certifying the quality of their investees in the going public process, despite the higher Legality index in the US (see Table 2). It is also noteworthy that the US-based investee companies have had a high proportion of write-offs.

3.2. Univariate Comparison Tests

Table 3 provides summary statistics and comparison of proportions tests for different variables of interest (as discussed, each variable is defined in Table 1). The cut-off values for the comparison tests are provided at approximately the median values for each variable. The comparison of proportion tests are provided for the full sample including unexited investments, so that potential biases with the presence of exited versus unexited investments are accounted for in the tests (as done in the multivariate tests in the next section).

[Table 3 About Here]

Tests #1-3 in Table 3 indicate the proportion of each exit outcome for the Legality index above and below 20; Test #1 is for the full sample, Test #2 is for the subsample excluding the US; Test #3 is for the subsample excluding the US and Taiwan. Tests #1 and 2 indicate a greater proportion of write-offs for the countries with higher legality indices. This might be a little surprising, but perhaps reflects an increased willingness among VCs to pursue risky opportunities in countries with better legal protections. Test #3 indicates that IPO outcomes are more likely among countries with higher legality indices. Note that this is in contrast to Tests #1 and 2, where the US and Taiwan observations are included. In short, Legality does appear to matter, although Taiwan appears to be an outlier country with a high proportion of successful exits despite a low legality index, as does the US which has a low proportion of successful Asia-Pacific VC backed exits despite a high legality index. This issue of outlier countries in the data is considered in the multivariate analyses in the next section.

It is also noteworthy that a lower proportion of private exits is observed among countries with high legality indices among the subsample of countries excluding Taiwan and the US (Test #3). As discussed in section 2, we were unable to ascertain the specific type of private exit (acquisitions, secondary sales and buybacks) for many of the transactions. Hence, it is for this reason that the sample sizes are smaller for the specific types of private exits in Table 2. Based on the limited data from the subsample of known private exits, it appears that buybacks are more common among countries with worse legality indices. This is consistent with our expectations (see section 2), but additional work with more complete datasets on buybacks and other private exits could be fruitful.

Tests #4-6 in Table 3 provide comparisons of proportions of exits for different market capitalization values for the country-years in our dataset. We carry out these tests with the same

subsamples as in Tests #1-3 (the full sample, the subsample without the US, and the subsample without Taiwan). In contrast to the evidence in support of the role of legality in facilitating IPO exits, the data do not indicate that IPOs are more likely among countries with larger stock markets. Test #5 nevertheless indicates private exits and write-offs are statistically more likely in countries with smaller markets.

Finally, Tests #7-10 in Table 3 provide comparisons across specific countries. Tests #7-9 indicate IPOs are more likely in Taiwan and Australia than in the US, but the frequency is not materially different in Taiwan or Australia. The frequency of private exits is smaller in Taiwan relative to Asia-Pacific VC-backed companies (Test #7), and relative to all other countries excluding the US and Australia (Test #10). Likewise, IPOs are more likely in Taiwan relative to the sample of all other countries excluding the US and Australia (Test #10).

3.3. Correlation Matrix

A correlation matrix is provided in Table 4. The statistically significant coefficients (at the 5% level) are highlighted. The evidence highlights additional trends across the variables in the data. The correlations in Table 4 are provided for the subsample of exited investments only, to provide a companion perspective on the data relative to Table 3. A few additional variables are also considered, such as IRRs, which show IPOs as the most profitable exit outcomes (consistent with the picture in Figure 1). Other variables pertinent to VC fund characteristics and entrepreneurial firm characteristics are presented to indicate the apparent trends in the matrix (but not discussed at length for reasons of conciseness).

[Table 4 About Here]

The correlation matrix also reports components of the Legality index. The separate components of the Legality index are not studied individually in the regression analyses because they are highly correlated. As well, some individual legal indices might potentially be outliers with respect to the observations in the data, such as the dummy variable for civil law countries. For example, in the correlation matrix, IPOs appear more likely for civil law countries, but that result is influenced by the fact that for the US exits in our sample (a common law country) IPOs were much less likely, and for the Taiwan exits in our sample (a civil law country) IPOs were more likely (see Tables 2 and 3). Because we have data from only 468 investments and 333 exits, the weighted average Legality index mitigates the possibility of outlier observations with respect to particular countries' components of the Legality index. (More generally, there is no theory or legal reason why one component of the Legality index should be

more important for IPOs than another.) Note that we also checked for robustness by excluding certain outlier countries like the US and Taiwan in the multivariate empirical analyses (as in Table 3 discussed above).

The correlations also provide guidance as to potential collinearity problems for robustness checks in the multivariate tests in the next section. It indicates that variables other than Legality are also correlated with the exit choice. Most of these, among others not presented in Table 4, are included in the multivariate analyses below as control variables, such as those related to VC fund characteristics, entrepreneur and investment characteristics, and market conditions. Table 4 in fact indicates fund-specific factors are extremely important in driving the exit outcomes (i.e., that VC fund characteristics and fund manager abilities are important for achieving IPO exits, consistent with Gompers and Lerner, 1999). As such, we include such variables as controls in the multivariate empirical analyses below.

4. Multivariate Analyses of Exit Outcomes

In this section we present various logit models of the likelihood of an IPO exit, versus a private exit (an acquisition, secondary sale or buyback) or a write-off exit. Subsection 4.1 considers multinomial logit estimates for IPOs versus private exits versus write-offs. Subsection 4.2 reports the results of sample selection models in the spirit of Heckman (1976, 1979), which accounts for the non-randomness of exited versus unexited investments.

The sensitivity of the results to alternative specifications is rigorously considered, and a total of 16 different regression models are explicitly presented to show robustness. The regression tables report the marginal effects (only) to highlight economic significance alongside statistical significance (standard logit coefficients are nevertheless available upon request). Robustness checks not explicitly presented are discussed herein, and limitations and alternative explanations are considered in section 5.

4.1. Multinomial Logit Estimates

Table 5 presents six alternative multinomial logit models of IPOs versus private exits versus write-offs as a function of different right-hand-side variables (the variables were defined in Table 1 and discussed above). The specific types of private exits (acquisitions, secondary sales and buybacks) are not studied separately in subsection 4.1 because, as discussed in sections 2 and 3 above, there were a number of unknown types of private exits. The unexited investments are treated as a fourth possible outcome (to

account for the possible non-randomness in the exits observed).¹⁵ Model (1) in Table 5 expresses the exit outcomes as a function of legal and institutional conditions (the log of the legality index, the log of stock market capitalization, a dummy variable for US exits), market conditions (the log of MSCI returns 1 year and 1 month prior to exit), VC fund characteristics (a dummy variable for captive VCs, the log of fund capital per manager, the log of the years of experience of the VCs, and a dummy variable for sector specialization), and entrepreneurial firm and investment characteristics (the log of the firm's industry market/book ratio and the log of the size of the investment). Logs are used to account for the intuitive declining marginal impact of, for example, an improvement in Legality among changes at high values versus comparable changes at low values.¹⁶ Models (2) and (3) explicitly show the sensitivity of the results excluding the market capitalization variable (Model 2) and the Legality index (Model 3), because these variables are collinear (the correlation coefficient is 0.46; see Table 4). Models (4) – (6) in Table 5 add in extra variables, including dummy variables for other countries, exit year dummy variables for the Internet bubble, other proxies for VC fund quality, and other transaction specific variables. Other variables in our dataset (discussed above in section 3) were considered but immaterial to the analysis (but are nevertheless available upon request). The next subsections also present other specifications, as explicitly discussed below.

[Table 5 About Here]

The central theme in this paper is to study the role of Legality on the likelihood of IPO exits versus the other VC exit outcomes, and to compare the role of Legality to other variables such as stock market capitalization and investment specific variables. Table 5 provides the first set of key results in this regard. With some variation across the six models presented, the marginal effect is approximately 0.8, which indicates an increase in the Legality index from 20-21 increases the probability of an IPO by 1.7% [$(\text{Log}(21) - \text{Log}(20)) * 0.8$], whereas an increase in Legality from 10-11 increases the probability of an IPO by 3.3% [$(\text{Log}(21) - \text{Log}(20)) * 0.8$]. These core results are consistent with our central expectations (see subsection 2.1).

¹⁵ We do not present the marginal effects for non-exits in Table 5 for reasons of conciseness. The results are robust to simply dropping the non-exited outcomes from the sample. Alternative ways of dealing with non-exits are discussed and explicitly considered in the next subsections; see in particular subsection 4.2.

¹⁶ Model specification tests (such as the likelihood dominance criterion) showed either no difference between, or in some cases a slight preference for, the log specification. For example, Model (1) in Table 5 with Legality in levels (not logs) showed the marginal effect on Legality at 0.056 (and significant at the 1% level), and the loglikelihood function was -399.3109, thereby indicating an insignificant LR statistic comparing the reported model versus the one in logs. We chose to present only the log results as they more intuitively account for a declining marginal impact of Legality (among other variables) at higher levels.

Importantly, compare the results on the Legality in Table 5 variable to the variable for the stock market capitalization variable in the exit jurisdiction, both in the models where both variables are included (Models 1 and 4-6) and the models where the variables are included separately (Models 2 and 3). None of the models indicate an increase in the probability of an IPO that is directly associated with the size of a country's stock market (and this finding is consistent with our summary statistics presented in Tables 3 and 4). This is quite a significant finding because the conventional wisdom has been that stock markets by themselves facilitate venture capital markets. That proposition has merely been just that – a proposition (Black and Gilson, 1998) – and not something that has been empirically tested. Our results, by contrast, show that the mechanism by which IPOs are facilitated in connection with venture capital is by virtue of a high quality legal environment. It is of course the case that all three variables (venture capital IPO exits, Legality and market capitalization) are correlated. But again, Legality is the central mechanism that facilitates VC-backed IPOs, at least in our data, and not stock market activity by itself (as conjectured by Black and Gilson, 1998). Stock market capitalization only is indirectly relevant by virtue of being correlated with Legality.¹⁷

As indicated in the prior sections of this paper, many Asia-Pacific VCs seek to exit their investments in the United States.¹⁸ Our data indicates that this investment strategy less often results in IPO exits, and more often results in private exits. One likely explanation for this finding is that Asia-Pacific VCs do not have the same reputation that would facilitate IPO exits in the US, where such public offerings would be in competition with more reputable US-based VC funds (consistent with Barry *et al.*, 1990; Megginson and Weiss, 1991; Gompers and Lerner, 1999; and Hsu, 2004). Another possible explanation is that the geographic distance is a factor in hampering effective venture governance.¹⁹

A surprising finding is the inability of Asia-Pacific VCs to time markets in respect of public market conditions (the MSCI variables). Although many of the exits in our sample were in 2000 (Figure 1), many also occurred after the crash of the bubble in April 2000. This is suggestive of less ability

¹⁷ In Model (6) in Table 5, there appears to be even a negative association between stock market capitalization and the probability of a VC-backed IPO (consistent with the correlations in Table 4). That result, however, is not robust to the alternative specifications.

¹⁸ The empirical question of why Asia-Pacific VCs would want to do this (e.g., for improved laws, etc.) is a subject beyond the scope of this paper, but perhaps worthwhile for future research. Similar legal scholarship has been carried out in respect of the Israeli venture capital market in connection with the US (Rock, 2001).

¹⁹ However, controls for US offices and US experience (see section 3 for these variables), etc., were insignificant when included in the specifications.

among VCs in the Asia-Pacific to time the IPO market, particularly relative to US VCs (as documented in Lerner, 1994b; Gompers and Lerner, 1999).

Some of our proxies for VC skill show a relation between ability and alternative exit outcomes. First, note that captive VCs are approximately 8-9% more likely to experience write-offs in each of the 6 models, and 14% less likely to experience IPOs (although the latter result is only significant in Model 6). These findings are consistent with those in Gompers and Lerner (1999) for US VCs, in that they show captive funds typically face organizational problems that render them less suitable for successful VC investment (due to pay per performance, autonomy, conflicting objectives, etc.). Second, larger funds are more likely to obtain IPOs. This is expected, as more skilled VC managers attract more capital from investors, which in turn is related to a desire on behalf of higher ability entrepreneurial firms to seek financing from such VC funds (consistent with Hsu, 2004). Third, note that our proxy for years of experience is not significant in the ways in which we would predict (we would predict a positive association with IPOs). Nevertheless, the data are consistent with theoretical work (e.g., Kanninen and Keuschnigg, 2004), in that the data suggest it is difficult for VC managers to acquire skills that give rise to successful VC financing (i.e., simply putting in the years is insufficient). Fourth, it is noteworthy that funds with sector specialization were more likely to have IPOs and write-offs, and less likely to have private exits. These findings indicate sector focus gives rise to specialization and enhanced ability and facilitate successful outcomes, but also increases fund risk. Finally, note that our various additional proxies for VC skill (in Model 6) did not materially affect the other results. In regards to stage specialization (funds focused on early stages), such VCs experienced fewer IPO and more private exit outcomes (indicating risks and/or inability at more difficult tasks at funding nascent firms). For VCs that had previously financed more funds, write-offs were more likely (again somewhat surprising, but consistent with the number of years experience variable already mentioned).

In respect of controls for entrepreneurial firm and transaction specific factors, first note that firms with higher industry market/book ratios are more likely to go public (consistent with Gompers and Lerner's, 1999, seminal work on the US VC market). This is expected, as higher industry market/book ratios indicate higher growth options (in the spirit of Fama and French, 1992). Second, note that proxies for size (amount of VC investment) are more closely connected with private exits than IPOs (which is somewhat unexpected). Third, note that Models (4) – (6) also include syndication, co-investment and investment duration variables. While these latter variables are potentially endogenous, we did consider potential instruments (e.g., pertaining to country-specific factors), and did not find any materially differences to the coefficients reported and/or the primary variables of interest (such as Legality,

discussed above). We include these later variables to show robustness, and to provide a reasonably complete specification. There is some (albeit weak) evidence in Model (4) that syndication lowers the probability of a write-off (perhaps due to improved screening of low quality entrepreneurs; see Lerner, 1994a; Gompers and Lerner, 1999), and evidence in Model (6) that co-investment lowers the probability of a write-off (perhaps because the motive to co-invest is for one VC fund to bail out the prior bad investments of another VC fund in the same VC organization; see Gompers and Lerner, 1996, 1999). Longer investment duration is associated with a smaller probability of a write-off (consistent with the idea that plums take a long time to bring to fruition, but lemons ripen quickly, as discussed by Megginson and Weiss, 1991; Gompers and Lerner, 1999; Cumming and MacIntosh, 2003a,b).

Finally, note that where we have information on other variables that could have been included, they were considered but immaterial to the central results. There are also other variables in the dataset but could not be included in the multinomial logit regressions. For example, it is not possible to include dummy variables for other countries where all four outcomes (IPO, private exit, write-off and no exit) have not occurred (e.g., Taiwan and other country dummies, as well as exit year dummy variables). Some of those variables are nevertheless possible to consider in a binomial logit framework (not reported in this paper).

4.2. Sample Selectivity Corrected Estimates

This section presents a robustness check for the determinants of IPOs, with regard to statistical sample selection issues in the spirit of Heckman (1976, 1979) (and more specifically, based on methods employed in Wynand and van Praag, 1981). We consider determinants of IPOs, with first considering whether or not there has been an exit of any type. The non-randomness of observing an exit may have resulted in a bias in the above-reported results (for statistical reasons first explained by Heckman), and hence it is worthwhile to present this robustness check.

Our specification first considers the determinants of an exit (versus no exit) as a function of the year of first VC investment (dummy variables for different investment years for which there have and have not been exits), as well as country-specific stock market capitalization (to account for the possibility that exit itself is ‘easier’ among countries with bigger markets; our first step specification also drops this latter variable as a robustness check). Our second step model includes many of the core variables that were used in subsection 4.1. For reasons discussed in Puhani (2000), we avoid using the same variables in the first and second step regressions to mitigate the sensitivity of the specifications to the included variables

(which means that we exclude market capitalization in the second step regression in cases where it was included in the first step regression, and we do not include exit year dummies because they are highly correlated with the investment year dummies). We present somewhat models with a more concise set of right-hand-side variables (relative to, for example, Model (6) in Table 5) in order to mitigate problems of collinearity across variables and to limit sensitivity of results to included variables in the first- and second-step regressions. We present four alternative two-step models (7—10) in Table 6 to show robustness.

[Table 6 About Here]

The Step 1 regressions in Table 6 (Models 7—9) indicate exit is about 24% less likely (based on unconditional probabilities) in the data if the first investment was in 1999, and 4% less likely if the first investment was in 1998. The Step 1 regression in Model (10) also includes a variable for stock market capitalization to account for the possibility of differing non-exits across countries in the data. Our data (see Table 2) reflect fewer non-exits in countries with smaller stock markets.

The Step 2 regressions consistently show an increase in the probability of an IPO exit for VCs in countries with higher legality indices, and this effect is significant at the 5% level in Model (7) and at the 1% level in Models (8) – (10). The economic significance is slightly less than that reported above in the context of the multinomial logit models in Table 5. By contrast, as before, stock market capitalization does not directly impact upon the IPO exit frequency.

The US dummy variable is also negative and significant in each of the models in Table 6, and the Taiwan dummy variable is insignificant, consistent with the results reported in Table 5. The MSCI market conditions variables were also insignificant, as before. Note that the inclusion of other variables (such as a dummy variable for Australia, and/or exit year dummy variables) gives rise to an over-specification problem with these sample selection models due to collinearity as to the investment year and other variables (Puhani, 2000, discusses the high degree of sensitivity of the results of sample selection models to this collinearity problem with the first and second stage regressions). As such, we did not include those extra variables in any of the specifications in Table 6.

Finally, note that one of the proxies for VC fund ability (fund size per manager) in Table 6 is positive and significant, consistent with the results in Table 5, consistent with other work on VC sophistication (most recently, see Hsu, 2004). The results in Table 6 are also consistent with the notion that

firms higher market/book industries are more likely to go public (consistent with Gompers and Lerner, 1999), as expected for reasons discussed above in conjunction with Table 5.

5. Limitations, Alternative Explanations and Future Research

This research would not have been possible without the proprietary data provided by Wilshire Associates, Inc. The data significantly extends prior knowledge of venture finance in the Asia-Pacific, and provides a first-ever test of the role of Legality in exits. There is always scope for expanding the breadth of data and depth of detail in VC data because empirical VC research invariably involves the use of confidential information. As such, as with most topics and papers on VC, there are limitations, alternative explanations and ample room for future research.

First, we note that there are some variables that we have in the data (for instance, some are reported in Tables 3 and 4), but were immaterial to the multivariate analyses and hence not included. Similarly, alternative types of variables could have been used, such as specific industry dummy variable instead of the industry market/book variable. Regardless, the use of different types of substitute variables did not materially affect the nature of the results presented and the inferences that can be drawn from the data.

Second, note that in our empirical specifications we could have excluded certain outlier countries from the sample (as done in one of the robustness checks in an earlier version of this paper). The main outlier countries in our sample include Taiwan (which has had a successful VC exit market despite a comparatively low Legality index) and the US (that is, Asia-Pacific backed US firms) (which has had unsuccessful exits despite a comparatively high Legality index). Excluding those outlier countries tends to strengthen our core results pertaining to the connection between Legality and IPOs. Further work might nevertheless explore why VCs invest and exit abroad (in the spirit of Rock's 2001 descriptive analysis of Israeli VC funds coming to the US).

Third, additional details could be obtained in regards to contractual differences among VCs in the Asia-Pacific. Our analysis is limited to the effect of syndication and co-investment on exit outcomes. We acknowledged some potential endogeneity issues in respect of those variables, and perhaps unsuitable instruments (various ones that we considered did not materially impact the results presented, so we only presented the results without instruments, and results without the presence of potentially endogenous variables).

Fourth, other contractual terms that we do not know about in our dataset may impact exit outcomes, particularly if they affect control over exit. In an analysis of European contracts and exits, for example, Cumming (2002) finds a strong connection between the use of strong VC control rights and the probability of acquisition exits. The sample of countries considered by Cumming (2002), however, is much more homogenous (continental Western Europe, and a few Eastern Europe exits). The Asia-Pacific countries, by contrast, are more segmented and involve much more pronounced differences in Legality, and average Legality index values in the Asia-Pacific are much lower relative to the sample of predominantly Western European countries studied by Cumming (2002). Hence, Legality likely has a more pronounced role among the countries that we study, relative to more homogenous samples where specific features of the investment process are more likely to play a more pronounced role in the ways in which companies are exited.

If contractual structures did matter to exit outcomes, based on Lerner and Schoar (2004b) we may infer that VCs in countries with higher Legality indices (at least among countries in our comparable sample) would be more likely to use contracts that give rise to larger investments with higher valuations (and hence increase the probability of an IPO, based on the information in the data that we do have pertaining to size and other proxies). Similarly, Cumming *et al.* (2004) find evidence that Legality is positively associated with better governance structures (more syndication, less co-investment, more board seats, more high-tech investments), which are all positively associated with IPOs in our dataset herein. In effect, Legality itself appears to be the source of international differences in contracts and governance, which facilitates IPO exit outcomes. We therefore do not have reason to believe that missing investment details would materially change our results on the central role of Legality in fostering VC-backed IPOs. Future work on these lines may nevertheless uncover something that we do not have access to at this time.

Our results and analyses are of course confined to the sample of countries studied in the Asia-Pacific. Other studies of more homogenous countries with respect to Legality (e.g., within continental Europe; see Cumming, 2002) may not find Legality to be an important factor that explains VC-backed IPO exits. The vast differences in Legality across the Asia-Pacific countries we study in this paper are important for the results observed. Data derived from countries with little differences in legal indices are unlikely to yield the same results documented herein.

The analyses in this paper also suggest various directions for future research on VC in the Asia-Pacific. For example, the role of tax laws on the amount of venture finance in Asia-Pacific markets

(based on the theoretical work of Kanninen and Keuschnigg, 2003a,b; see also Keuschnigg, 2003, 2004, and Keuschnigg and Nielsen, 2001, 2003a,b,c, 2004) is also worthy of empirical study (see also Lerner, 1999, 2002). The influence of Asia-Pacific VCs on rates of innovation among entrepreneurial firms is unknown (see Kortum and Lerner, 2000, for US evidence). Venture capitalist value-added activities in Asia-Pacific could be analyzed (see, e.g., Hsu, 2004, for a recent innovative US study). Some of our results were suggestive (although not completely conclusive) of a comparative lack of skill among VCs in the Asia-Pacific, at least relative to the US. Equally interesting would be further study of Asia-Pacific VC fundraising and sources of capital, as well as the short- and long-run IPO performance among Asia-Pacific VC-backed companies (see, e.g., Gompers and Lerner, 1999, for US evidence; see also Hamao *et al.*, 2000, for evidence from Japan, and Da Silva Rosa, 2003, for Australian evidence, and Lee, 2004, for Korean evidence), and the valuation of private equity investments in the Asia-Pacific (see, e.g., Gompers and Lerner, 2000, for US evidence). This type of research would help in better understanding exits and returns in Asia-Pacific venture capital and private equity markets.

7. Concluding Remarks

This paper provides the first cross-country empirical analysis of the relation between Legality and VC exits outcomes: IPOs, private exits (acquisitions, secondary sales, buybacks) and write-offs. With a sample of 468 VC investments in Asia-Pacific countries, we show an increase in the Legality index from 20-21 (e.g., approximately the difference between Hong Kong and Australia) increases the probability of an IPO by about 1.7%, whereas an increase in Legality from 10-11 (e.g., approximately the difference between the Philippines and Indonesia) increases the probability of an IPO by 3.3%.

In demonstrating this central role of Legality in VC exits, we controlled for many factors, including market conditions, country-specific factors such as moving to the US, VC fund characteristics, as well as entrepreneurial firm and deal-specific characteristics. Our findings on a number of these controls are consistent with prior work. For example, our data are consistent with the role of the ability of the VC (consistent with Hsu, 2004) and the size and industry market/book value of the entrepreneurial firm (consistent with a large number of seminal studies collected in the Gompers and Lerner, 1999, book).

It is particularly noteworthy that none of our summary statistics and/or multivariate models indicated an increase in the probability of an IPO that was directly associated with the size of a country's stock market (although the size of a country's stock market is strongly and positively associated with Legality). This complementary finding in our paper is quite significant, because the conventional wisdom

has been that stock markets by themselves facilitate venture capital markets (Black and Gilson, 1998). That proposition, however, has not previously been empirically tested. Our data show that the mechanism by which IPOs are facilitated in connection with venture capital is by virtue of a high quality legal environment. The degree to which a country's stock market is active is nevertheless indirectly relevant by virtue of being correlated with Legality. Our dataset, therefore, suggests a restatement of the widely recognized Black and Gilson (1998) hypothesis pertaining to active stock markets and active venture capital markets, as follows: high quality legal environments facilitate successful exit environments and active venture capital markets, and these variables happen to be correlated with active stock markets. As we have discussed, we hope that further work will explore the robustness of our findings to other international contexts, and with additional details pertaining to the structure of VC transactions and the governance that VCs provide to their investees.

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Figure 1 . Investment and Exit Years in the Dataset

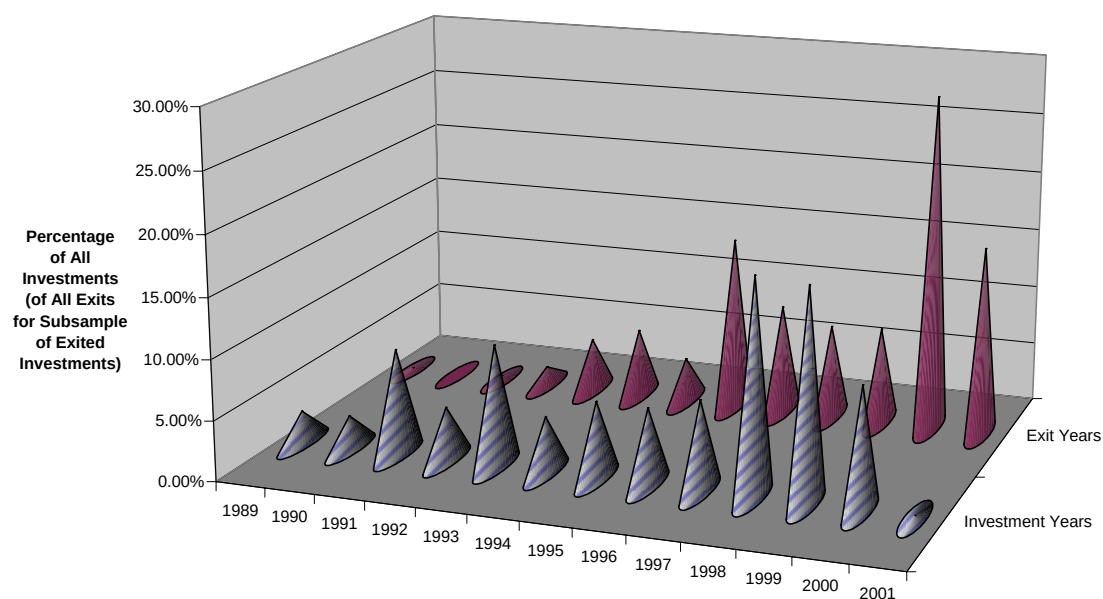


Figure 2. Frequency of and Returns to Alternative Exit Outcomes

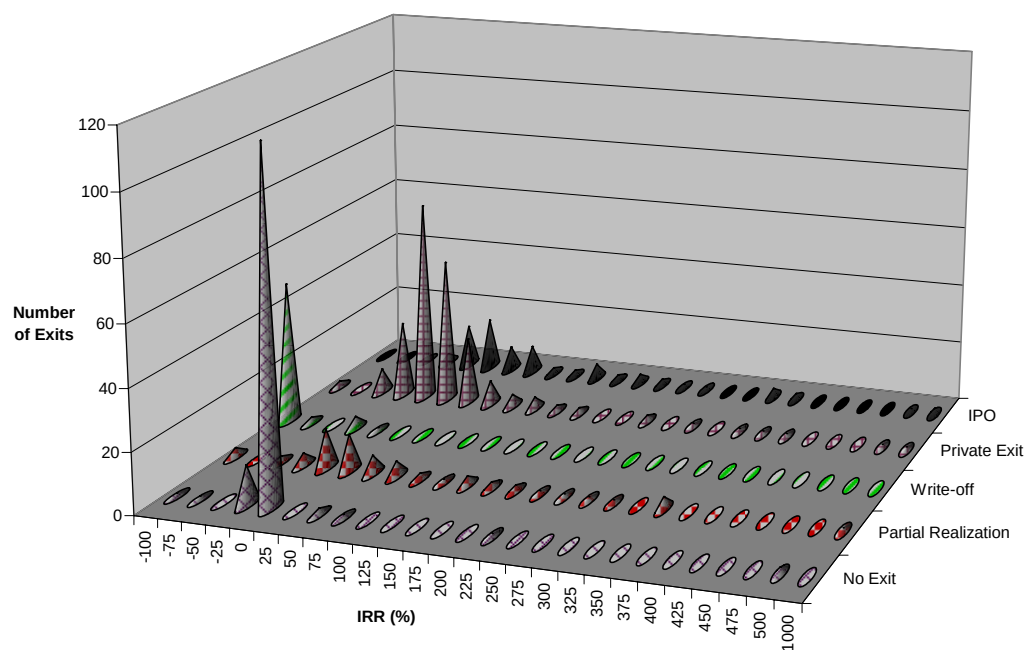


Table 1. Definition of Variables	
This table defines the variables used in the subsequent tables.	
Variable Name	Description
IPO	A dummy variable equal to 1 if the VC exited by means of an IPO exit.
Acquisition	A dummy variable equal to 1 if the VC exited by means of an acquisition (trade sale) exit (i.e., both the VC and entrepreneur sell to another company).
Secondary Sale	A dummy variable equal to 1 if the VC exited by means of a secondary sale exit (i.e., the VC sells his stake, but the entrepreneur retains her stake).
Buyback	A dummy variable equal to 1 if the VC exited by means of a buyback exit (entrepreneur repurchases from the VC).
Private Exit	A dummy variable equal to 1 if the VC exited by means of a private exit (either an acquisition, secondary sale, or a buyback).
Write-off	A dummy variable equal to 1 if the VC exited by means of an IPO exit.
IRR	The VC's internal rate of return on the investment.
Legality	The weighted average of following factors (based on Berkowitz et al. (2003)): civil versus common law systems, efficiency of judicial system, rule of law, corruption, risk of expropriation, risk of contract repudiation, shareholder rights (as in La Porta et al., 1997, 1998). Higher numbers indicate 'better' legal systems.
Market Cap	The country-specific stock market capitalization (in billions of 2001 US dollars).
MSCI 1 Year (or 1 Month) Prior	The MSCI index is the country-specific Morgan Stanley Capital International Stock Market Index returns; variables for returns for the 1-month prior to exit and 1-year prior to exit are explicitly indicated.
Exit Year Dummy Variables	Dummy variables for exit years (particularly 1998 - 2001).
Exit Country Dummy Variables	Country dummy variables (specifically, the US, Australia and Taiwan).
Captive VC	A dummy variable equal to 1 if the VC was a bank or corporate captive VC (and 0 if the VC was a limited partnership).
Fund Size / Manager	The capital raised by the fund, divided by the number of fund managers.
Portfolio Size / # Managers	The number of entrepreneurial firms in the portfolio, divided by the number of fund managers.
Fund Number	The number of funds raised by the VC organization at the time of investment in the firm.
Years Experience	The average number of years of VC experience at time of investment (robustness considered for number of years of experience at time of exit as well).
US VC Experience	A dummy variable equal to 1 if the VC had US experience.
Stage Specialization	A dummy variable equal to 1 if the VC focused on entrepreneurial firms in their early stage of development.
Sector Specialization	A dummy variable equal to 1 if the VC focused on a particular industry.
Industry Market / Book	The investee firms' industry market/book ratio.
Size of Investment	The size of the VC's investment (in millions of 2001 US dollars).
High-Tech Entrepreneur	A dummy variable equal to 1 if the entrepreneur was in a high-tech industry (robustness to different types of industries was considered).
Early or Expansion Stage	A dummy variable equal to 1 if the entrepreneur was in the early or expansion stage at the time of first VC investment (more specific stage classifications were not possible for the countries in the data).
Syndication	A dummy variable equal to one if there was more than 1 VC fund from different VC organizations that financed the entrepreneurial firm.
Co-Investment	A dummy variable equal to one if there was more than 1 VC fund within the same VC organization that financed the entrepreneurial firm.
Duration of Investment	The number of months from the date of first VC investment to the date of exit (or to December 2001 if unexited).

Table 2. Exits Data for the Countries Represented in the Exits Database, and the Legality Components¹

This table presents the legality indices, the average country stock market capitalization values (weighted for our exit years in each country), the average and median IRRs by country of exit, and the number of exits of each type in our dataset. Private exits encompass acquisitions, secondary sales and buyback exits, as defined in Table 1. The fewer cases of known private exits are also indicated.

Country	Common Law / Civil Law (Civil=1)	Efficiency of Judicial System	Rule of Law	Corruption	Risk of Expropriation	Risk of Contract Repudiation	Shareholders Rights ¹	Legality ²	Total Market Capitalization (Billions of 2001 US\$) ³	Average IRR in our Dataset (Full Exits Only)	Median IRR in our Dataset (Full Exits Only)	# IPOs in our Dataset	# Private Exits in our Dataset	# Known Acquisitions in our Dataset	# Known Secondary Sales in our Dataset	# Known Buybacks in our Dataset	# Write-offs in our Dataset	# Partial Exits in our Dataset	# Unexited Investments in our Dataset
New Zealand	0	10	10	10	9.69	9.29	4	21.55	19.69	0.905	0.365	4	10	7	2	1	0	1	1
Asia-Pacific VC Investments in US ⁴	0	10	10	8.63	9.98	9	5	20.85	12380.03	0.480	0.064	9	49	3	0	2	21	9	38
Australia	0	10	10	8.52	9.27	8.71	4	20.44	242.26	0.164	0.189	27	43	18	5	9	23	15	22
Singapore	0	10	8.57	8.22	9.3	8.86	4	19.53	130.42	0.341	0.145	1	7	0	0	0	3	1	2
Hong Kong	0	10	8.22	8.52	8.29	8.82	5	19.11	278.95	-0.056	0.070	2	11	1	0	0	3	3	8
Malaysia	0	9	6.78	7.38	7.95	7.43	4	16.67	108.65	-0.168	-0.168	0	1	0	0	0	0	0	0
India	0	8	4.17	4.58	7.75	6.11	5	12.80	213.19	1.020	1.020	2	1	0	0	1	0	1	0
Thailand	0	3.25	6.25	5.18	7.42	7.57	2	12.94	70.02	-0.101	-0.101	0	9	1	0	0	3	0	0
China ⁵	1	8.54	8.68	8.03	9.45	9.47	2.33	19.23	461.42	-0.063	0.084	2	12	0	0	0	2	2	2
Taiwan	1	6.75	8.52	6.85	9.12	9.16	3	17.62	388.13	0.723	0.445	31	39	1	1	2	0	31	62
South Korea	1	6	5.35	5.3	8.31	8.59	2	14.23	260.00	0.145	0.250	0	3	0	0	0	0	0	0
Indonesia	1	2.5	3.98	2.15	7.16	6.09	2	9.16	25.68	-0.029	-0.015	0	14	0	0	1	0	1	0
Philippines	1	4.75	2.73	2.92	5.22	4.8	3	8.51	13.04	-0.003	-0.003	0	1	0	0	0	0	0	0

1. The Shareholder Rights Index is derived from the sum of dummy variables on shareholders rights from La Porta et al. (1998, pp. 1130-1131).

2. The Legality Index is constructed from Berkowitz et al. (2003): $\text{Legality} = 0.381 * (\text{Efficiency of Judiciary}) + 0.5778 * (\text{Rule of Law}) + 0.5031 * (\text{Corruption}) + 0.3468 * (\text{Risk of Expropriation}) + 0.3842 * (\text{Risk of Contract Repudiation})$

3. Source: Dow Jones. Values are expressed for average amounts in specific exit years from 1989-2001 in our dataset.

4. The "U.S. investments" in the database are Asia-Pacific venture capitalists' exits among companies located in the U.S.

5. We have employed the average German legal origin values for China. These values are consistent with, for example, the Rule of Law values for China as used by Durnew and Kim (2004). Our China values are also somewhat consistent with the Allen *et al.* (2002) legal index estimates for China, but perhaps the average German legal origin values over-estimate the legal indices for China relative to that reported by Allen *et al.*; regardless, adjustments downwards on the China legal index did not affect our empirics.

Table 3. Difference Tests and Summary Statistics

This table presents summary statistics for Asia-Pacific VC-backed entrepreneurial firms by the proportion of exit vehicles: IPOs, acquisitions, secondary sales, buybacks and write-offs. Statistics are also presented for "private exits", which collectively refers to acquisitions, secondary sales and buybacks (the specific form of private exit was unknown in 145 cases). Note that the proportion of exits does not sum to 1.0, because there are a total of 135 unexited investments in the dataset. Statistics are presented for different values of the variables (cutoffs are approximately at the median levels) that are defined in Table 1. The tests are provided for the subsample of known acquisition, secondary sale and buyback exits for approximate guidance only; however, the smaller number of observations for those subsamples provides less statistical accuracy. *, **, *** Significant at the 10%, 5% and 1% levels, respectively.

Test #	Characteristic Tested	# Firms	Proportion of IPO Exits	Difference Test Statistic	Proportion of Private Exits	Difference Test Statistic	# Firms for which Type of Private Exit Known	Proportion of Acquisition Exits	Difference Test Statistic	Proportion of Secondary Sale Exits	Difference Test Statistic	Proportion of Buyback Exits	Difference Test Statistic	Proportion of Write-off Exits	Difference Test p-value
	Full Sample	468	0.167		0.427		55	0.564		0.145		0.291		0.118	
1	Average Legality Index of Jurisdiction of Exit > 20 (Including All Countries)	249	0.161	-0.278	0.418	-0.447	47	0.596	1.156	0.149	0.125	0.255	-1.287	0.177	2.873***
	Average Legality Index of Jurisdiction of Exit < 20 (Including All Countries)	219	0.174		0.438		8	0.375		0.125		0.500		0.050	
2	Average Legality Index of Jurisdiction of Exit > 20 (Excluding US)	132	0.235	1.113	0.417	-0.394	42	0.595	1.143	0.167	0.216	0.238	-1.369	0.174	2.364**
	Average Legality Index of Jurisdiction of Exit < 20 (Excluding US)	219	0.174		0.438		8	0.375		0.125		0.500		0.050	
3	Average Legality Index of Jurisdiction of Exit > 20 (Excluding US and Taiwan)	132	0.235	2.303**	0.417	-3.458***	42	0.595	0.364	0.167	Not Applicable	0.238	-1.009	0.174	0.693
	Average Legality Index of Jurisdiction of Exit < 20 (Excluding US and Taiwan)	87	0.080		0.655		4	0.500		0.000		0.500		0.126	
4	Average Market Cap. in Country of Exit > 300 (Including All Countries)	271	0.162	-0.218	0.369	-2.962***	9	0.444	-0.782	0.111	-0.225	0.444	1.010	0.092	-1.285
	Average Market Cap. in Country of Exit < 300 (Including All Countries)	197	0.173		0.508		46	0.587		0.152		0.261		0.152	
5	Average Market Cap. in Country of Exit > 300 (Excluding US)	154	0.227	1.017	0.331	-3.284***	4	0.250	-1.310	0.250	0.376	0.500	0.922	0.026	-2.716***
	Average Market Cap. in Country of Exit < 300 (Excluding US)	197	0.173		0.508		46	0.587		0.152		0.261		0.152	
6	Average Market Cap. in Country of Exit > 300 (Excluding US and Taiwan)	22	0.182	0.082	0.545	0.337	0	0.000	Not Applicable	0.000	Not Applicable	0.000	Not Applicable	0.182	0.263
	Average Market Cap. in Country of Exit < 300 (Excluding US and Taiwan)	197	0.173		0.508		46	0.587		0.152		0.261		0.152	
7	US Only	117	0.077	-2.574**	0.419	1.944*	5	0.600	1.123	0.000	Not Applicable	0.400	-0.316	0.179	P-Value < 0.00
	Taiwan	132	0.235		0.295		5	0.250		0.250		0.500		0.000	
8	US Only	117	0.077	-2.488**	0.419	0.684	5	0.600	0.156	0.000	Not Applicable	0.400	0.494	0.179	-0.312
	Australia	115	0.235		0.374		32	0.563		0.156		0.281		0.200	
9	Taiwan	132	0.235	0.001	0.295	-1.230	5	0.250	-1.314	0.250	0.390	0.500	0.913	0.000	P-Value < 0.00
	Australia	115	0.235		0.374		32	0.563		0.156		0.281		0.200	
10	Taiwan	132	0.235	1.990**	0.295	-5.675***	5	0.250	-1.537	0.250	0.412	0.500	1.111	0.000	P-Value < 0.00
	All Other Countries Excluding US, Australia	104	0.106		0.663		14	0.643		0.143		0.214		0.106	

Table 4. Correlation Matrix

This table presents correlation coefficients across selected variables, as defined in Table 1. Correlations are provided for the subsample of 333 exited investments only. Correlations greater than 0.11 in absolute value are significant at the 5% level and highlighted in underline font.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
(1)	IPO	1.00																						
(2)	Private Exit	<u>-0.68</u>	1.00																					
(3)	Write-off	<u>-0.24</u>	<u>-0.54</u>	1.00																				
(4)	IRR	<u>0.30</u>	0.01	<u>-0.36</u>	1.00																			
(5)	Log (Legality)	0.06	<u>-0.16</u>	<u>0.15</u>	0.04	1.00																		
(6)	Log (Market Cap)	-0.07	-0.08	<u>0.18</u>	0.02	<u>0.46</u>	1.00																	
(7)	Civil Law	<u>0.14</u>	0.08	<u>-0.26</u>	0.09	<u>-0.50</u>	<u>-0.24</u>	1.00																
(8)	Judicial	-0.01	<u>-0.14</u>	<u>0.19</u>	-0.01	<u>0.91</u>	<u>0.42</u>	<u>-0.69</u>	1.00															
(9)	Rule of Law	0.06	<u>-0.17</u>	<u>0.16</u>	0.04	<u>0.96</u>	<u>0.47</u>	<u>-0.51</u>	<u>0.85</u>	1.00														
(10)	Corruption	0.02	<u>-0.14</u>	<u>0.16</u>	0.02	<u>0.97</u>	<u>0.37</u>	<u>-0.61</u>	<u>0.93</u>	<u>0.93</u>	1.00													
(11)	Expropriation	0.03	<u>-0.12</u>	<u>0.11</u>	0.07	<u>0.88</u>	<u>0.66</u>	<u>-0.31</u>	<u>0.75</u>	<u>0.89</u>	<u>0.81</u>	1.00												
(12)	Repudiation	<u>0.11</u>	<u>-0.11</u>	0.02	0.10	<u>0.83</u>	<u>0.40</u>	-0.01	<u>0.56</u>	<u>0.77</u>	<u>0.76</u>	<u>0.79</u>	1.00											
(13)	Shareholder Rights	-0.07	<u>-0.09</u>	<u>0.20</u>	0.01	<u>0.67</u>	<u>0.62</u>	<u>-0.76</u>	<u>0.82</u>	<u>0.64</u>	<u>0.70</u>	<u>0.60</u>	<u>0.28</u>	1.00										
(14)	Captive VC	<u>-0.12</u>	<u>0.14</u>	-0.05	-0.06	<u>-0.36</u>	<u>-0.29</u>	0.09	<u>-0.26</u>	<u>-0.40</u>	<u>-0.30</u>	<u>-0.47</u>	<u>-0.32</u>	<u>-0.28</u>	1.00									
(15)	Log (Fund Size / Manager)	<u>0.36</u>	<u>-0.19</u>	<u>-0.17</u>	<u>0.18</u>	<u>-0.15</u>	-0.10	<u>0.47</u>	<u>-0.29</u>	<u>-0.22</u>	<u>-0.21</u>	<u>-0.11</u>	<u>0.13</u>	<u>-0.32</u>	0.07	1.00								
(16)	Log (Portfolio Size / # Managers)	<u>0.34</u>	<u>-0.22</u>	-0.10	<u>0.17</u>	-0.02	<u>0.27</u>	<u>0.31</u>	<u>-0.16</u>	-0.02	-0.10	<u>0.12</u>	<u>0.19</u>	<u>-0.04</u>	<u>-0.17</u>	<u>0.63</u>	1.00							
(17)	Log (Fund Number)	0.07	-0.10	0.05	0.04	<u>-0.12</u>	0.00	<u>0.23</u>	<u>-0.17</u>	<u>-0.15</u>	<u>-0.15</u>	-0.06	0.04	<u>-0.21</u>	-0.01	<u>0.32</u>	<u>0.14</u>	1.00						
(18)	Log (Years Experience)	<u>-0.13</u>	-0.06	<u>0.23</u>	-0.08	<u>0.11</u>	<u>0.15</u>	0.03	0.10	<u>0.13</u>	0.07	<u>0.16</u>	0.08	<u>-0.01</u>	<u>-0.32</u>	-0.10	<u>-0.28</u>	<u>0.19</u>	1.00					
(19)	US VC Experience	0.01	0.00	-0.01	-0.06	<u>-0.05</u>	<u>0.41</u>	<u>0.15</u>	<u>-0.11</u>	-0.07	<u>-0.11</u>	<u>0.15</u>	0.09	0.07	-0.09	<u>0.15</u>	<u>0.43</u>	<u>0.43</u>	-0.05	1.00				
(20)	VC Stage Specialization	<u>-0.24</u>	<u>0.12</u>	<u>0.12</u>	-0.05	<u>0.32</u>	<u>0.36</u>	<u>-0.22</u>	<u>0.30</u>	<u>0.35</u>	<u>0.31</u>	<u>0.44</u>	<u>0.23</u>	<u>0.36</u>	<u>-0.27</u>	<u>-0.12</u>	0.02	<u>-0.22</u>	0.09	0.02	1.00			
(21)	VC Sector Specialization	<u>0.20</u>	<u>-0.29</u>	<u>0.15</u>	0.00	<u>0.27</u>	<u>0.56</u>	<u>-0.12</u>	<u>0.24</u>	<u>0.28</u>	<u>0.20</u>	<u>0.41</u>	<u>0.22</u>	<u>0.39</u>	<u>-0.38</u>	-0.02	<u>0.36</u>	0.08	-0.07	<u>0.54</u>	<u>0.26</u>	1.00		
(22)	Industry Market / Book	<u>0.24</u>	<u>-0.24</u>	0.04	<u>0.19</u>	<u>0.14</u>	<u>0.18</u>	0.02	0.07	<u>0.14</u>	0.10	<u>0.18</u>	<u>0.18</u>	<u>0.14</u>	<u>-0.17</u>	<u>0.19</u>	<u>0.23</u>	-0.10	-0.03	<u>0.11</u>	0.00	<u>0.25</u>	1.00	
(23)	Size of Investment	-0.02	0.05	-0.04	-0.03	<u>-0.14</u>	<u>-0.20</u>	<u>0.16</u>	<u>-0.11</u>	<u>-0.23</u>	<u>-0.12</u>	<u>-0.21</u>	-0.04	<u>-0.19</u>	<u>0.28</u>	<u>0.34</u>	<u>-0.16</u>	<u>0.23</u>	<u>0.11</u>	-0.02	<u>-0.18</u>	<u>-0.22</u>	-0.07	1.00
(24)	Syndication	0.02	0.00	-0.03	<u>0.11</u>	-0.04	-0.03	<u>0.12</u>	-0.06	-0.03	-0.05	-0.01	0.01	-0.08	<u>0.18</u>	0.02	-0.08	<u>0.12</u>	0.10	<u>-0.12</u>	-0.01	<u>-0.13</u>	-0.02	0.05

Table 5. Multinomial Logit Regression Analyses of Exit Outcomes

This table presents multinomial logit estimates of the probability of IPO, private exit, and write-off exit outcomes as a function of legal conditions, VC fund characteristics, and market conditions. A fourth possible outcome -- no exit -- is also considered in the presented specifications (but not reported for reasons of conciseness). The numbers presented are the marginal effects, not the multinomial logit coefficients, in order to explicitly indicate the economic significance of the results. There were 11 observations skipped from the full sample of 468, due to a few missing or uncertain values with some of the variables for those observations. Variables are as defined in Table 1. White's (1980) robust standard errors are used. *, **, *** significant at the 10%, 5% and 1% levels, respectively.

	Model (1)			Model (2)			Model (3)		
Explanatory Variables	IPO	Private Exit	Write-off	IPO	Private Exit	Write-off	IPO	Private Exit	Write-off
Constant	-2.963**	6.362***	0.023	-2.834***	5.918***	-0.040	-0.283	2.364***	-0.371**
<u>Legal and Institutional Conditions</u>									
Log (Legality Index of Entrepreneur at Exit)	0.854**	-1.389***	-0.150	0.823***	-1.298***	-0.134			
Log (Market Capitalization in Country of Exit)	-0.0001	-0.058	-0.011				-0.021	-0.052	-0.002
Dummy Variable for US Exit of Asia-Pacific VC	-0.366**	0.582**	0.111	-0.357***	0.340***	0.065	-0.237	0.423	0.073
<u>Market Conditions</u>									
Log (1 + MSCI) in Year Prior to Exit	0.216	-0.895***	0.352**	0.178	-1.022***	0.327**	0.164	-0.880***	0.306**
Log (1 + MSCI) in Month Prior to Exit	-1.587**	-0.239	0.268	-1.608**	-0.409	0.208	-1.540**	-1.049	0.174
<u>VC Fund Characteristics</u>									
Captive VC	-0.030	-0.018	0.089**	-0.025	-0.018	0.087**	-0.030	-0.020	0.080**
Log (Fund Capital / Fund Manager)	0.081**	-0.217***	-0.051**	0.074**	-0.242***	-0.055***	0.049	-0.217***	-0.053***
Log (Years Experience at the Time of Investment)	-0.052	-0.283***	0.157***	-0.053	-0.313***	0.147***	-0.046	-0.337***	0.131***
Sector Specialist Dummy Variable	0.181***	-0.374***	0.097**	0.174***	-0.387***	0.092**	0.184***	-0.396***	0.077*
<u>Entrepreneur and Investment Characteristics</u>									
Log (Industry Market / Book Ratio)	0.080**	-0.070	0.008	0.079**	-0.063	0.008	0.095***	-0.069	0.007
Log (Book Value of Investment)	-0.008	0.015*	-0.002	-0.008	0.016**	-0.002	-0.010	0.019**	-0.001
<u>Model Diagnostics</u>									
Total Number of Observations		457			457			457	
Pseudo R ²		0.311			0.307			0.292	
Loglikelihood Function		-399.747			-402.017			-410.731	
Chi-squared statistic		360.175			355.636			338.206	

Table 5 continues on the following page...

Table 5. (Continued)									
Explanatory Variables	Model (4)			Model (5)			Model (6)		
	IPO	Private Exit	Write-off	IPO	Private Exit	Write-off	IPO	Private Exit	Write-off
Constant	-3.000**	6.436***	0.038	-3.195***	5.560***	-0.319	-1.914*	2.686**	-0.772
<u>Legal and Institutional Conditions</u>									
Log (Legality Index of Entrepreneur at Exit)	0.866**	-1.414***	-0.155	0.869**	-1.258***	-0.065	0.638**	-0.638*	0.001
Log (Market Capitalization in Country of Exit)	0.0029	-0.055	-0.015	0.006	-0.039	-0.015	-0.095*	0.088	0.006
Dummy Variable for U.S. Exit of Asia-Pacific VC	-0.375**	0.566**	0.133	-0.407**	0.504**	0.118	0.218	-0.283	0.066
Dummy Variable for Australia Exit							0.399***	-0.455***	0.056
<u>Market Conditions</u>									
Log (1 + MSCI) in Year Prior to Exit	0.222	-0.881***	0.325**	0.281	-0.780***	0.296**	-0.187	-0.102	0.289*
Log (1 + MSCI) in Month Prior to Exit	-1.642**	-0.292	0.367	-1.466*	-0.068	0.630	-1.264	0.108	1.156**
Year 2000 Dummy Variable							0.521***	-0.592***	0.071
Year 1999 Dummy Variable							0.205*	0.089	-0.295**
<u>VC Fund Characteristics</u>									
Bank VC	-0.040	-0.021	0.104***	-0.034	-0.017	0.101**	-0.144*	0.010	0.134***
Log (Fund Capital / Fund Manager)	0.078**	-0.223***	-0.044*	0.094***	-0.185***	-0.019	0.142***	-0.106*	-0.035
Log (Fund Number)							-0.015	-0.143	0.158***
Log (Years Experience at the Time of Investment)	-0.064	-0.287***	0.172***	-0.042	-0.221***	0.172***	-0.130*	-0.014	0.144***
Sector Specialist	0.179***	-0.373***	0.099**	0.212***	-0.364***	0.124***	0.143	-0.247**	0.104*
Stage Specialist							-0.285***	0.250***	0.035
<u>Entrepreneur and Investment Characteristics</u>									
Industry Market / Book Ratio	0.082**	-0.069	0.005	0.085**	-0.054	-0.005	0.035	-0.054	0.018
Book Value of Investment	-0.008	0.015*	-0.003	-0.008	0.014	-0.004	-0.005	0.009	-0.004
Syndication	0.073	0.015	-0.084*	0.098	-0.064	-0.043	0.132	-0.094	-0.038
Co-Investment				-0.063	0.194**	-0.068	-0.070	0.182	-0.112*
Investment Duration				0.001	0.0001	0.001**	-0.001	-0.001	0.002***
<u>Model Diagnostics</u>									
Total Number of Observations		457			457			457	
Pseudo R ²		0.314			0.352			0.495	
Loglikelihood Function		-397.628			-375.493			-292.697	
Chi-squared statistic		364.413			408.684			574.276	

Table 6. Sample Selection Logit Regression Analyses of Exit Outcomes

This table presents bivariate logit estimates of the probability of IPO as a function of legal conditions, VC fund characteristics, and market conditions. Sample selection methods are used, in the spirit of Heckman (1976, 1979), whereby the first step regressions estimate the probability that there was an exit (as opposed to an on-going investment). The numbers presented for the Step 2 regressions are the conditional marginal effects (Partial derivatives of $E[IPO|Exit=1]$ with respect to the vector of characteristics), not the logit coefficients, in order to explicitly indicate the economic significance of the results. The numbers presented for the Step 1 regressions are the logit coefficients (conditional probabilities are not of interest in the first step). There were 11 observations skipped from the sample of exited investments, due to a few missing or uncertain values with some of the variables for those observations. Variables are as defined in Table 1. White's (1980) robust standard errors are used. *, **, *** significant at the 10%, 5% and 1% levels, respectively.

	Model (7)		Model (8)		Model (9)		Model (10)	
Explanatory Variables	Step 1: Dependent Variable: Exit=1	Step 2: Dependent Variable: IPO=1 Exit	Step 1: Dependent Variable: Exit=1	Step 2: Dependent Variable: IPO=1 Exit	Step 1: Dependent Variable: Exit=1	Step 2: Dependent Variable: IPO=1 Exit	Step 1: Dependent Variable: Exit=1	Step 2: Dependent Variable: IPO=1 Exit
Constant	0.694***	-2.439***	0.699***	-3.156***	0.697***	-3.088***	1.305***	-2.821***
<u>Legal and Institutional Conditions</u>								
Log (Legality Index of Entrepreneur at Exit)		0.567**		0.681***		0.704***		0.675***
Log (Market Capitalization in Country of Exit)				0.039		0.057	-0.094**	
Dummy Variable for U.S. Exit of Asia-Pacific VC		-0.215***		-0.409**		-0.535***		-0.332***
Dummy Variable for Exit in Taiwan						-0.094		-0.047
<u>Market Conditions</u>								
Log (1 + MSCI) in Year Prior to Exit						0.168		0.224
Log (1 + MSCI) in Month Prior to Exit						-0.885		-0.955
<u>VC Fund Characteristics</u>								
Captive VC						-0.114*		-0.085
Log (Fund Capital / Fund Manager)		0.170***		0.146***		0.128***		0.130***
Log (Years Experience at the Time of Investment)						-0.065		-0.047
<u>Entrepreneur and Investment Characteristics</u>								
Industry Market / Book Ratio				0.085***		0.079**		0.080**
Book Value of Investment		-0.019**		-0.014*		-0.010		-0.011
<u>Investment Year Dummy Variables</u>								
Invest in 1999	-0.771***		-0.684***		-0.685***		-0.681***	
Invest in 1998	-0.288*		-0.396*		-0.388*		-0.358*	
Invest in 1997	0.333		0.310		0.301		0.314	
<u>Model Diagnostics</u>								
# Observations where Dependent Variable=1	325	76	325	76	325	76	325	76
Total Number of Observations	459		459		459		459	
Loglikelihood Function	-404.511		-398.595		-394.501		-391.612	